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ACHTSAM study protocol: outreach diagnostics and assessment of tolerability in severe ME/CFS—a pilot study

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ABSTRACT

Introduction Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) is a severe, multisystem condition that often emerges after viral infections and affects physical and cognitive function. Severely affected patients are underrepresented in research due to immobility and exertional intolerance.

Methods This is a prospective, non-interventional observational study investigating the feasibility and tolerability of home-based diagnostics in patients with severe and very severe ME/CFS. Phase 1 includes remote identification using validated questionnaires. Phase 2 involves home visits for physiological, cognitive and biological assessments. The primary outcome is feasibility; secondary outcomes include tolerability and methodological barriers.

Ethics/dissemination The study protocol was approved by the Ethics Committee of the University of Freiburg (No. 25-1031-S1). Written informed consent is obtained from all participants. Results will be disseminated via peer-reviewed publications and patient support groups.

Trial registration number DRKS00035231; FRKS005506.

STRENGTHS AND LIMITATIONS OF THIS STUDY

- ⇒ Prospective, non-interventional observational design with home-based assessments reduces procedural burden and enables standardised data collection in patients unable to attend clinical settings.
- ⇒ A staged assessment protocol with rest periods and the option to pause or terminate procedures is used to minimise the risk of post-exertional malaise (PEM) during data collection.
- ⇒ Validated questionnaires and predominantly certified medical devices are employed to support standardisation and reproducibility of measurements across participants.
- ⇒ Subjective perceived exertion is assessed after each procedure, allowing systematic characterisation of individual tolerability limits to inform future study protocols.
- ⇒ Recruitment is geographically restricted, the sample size is small, and no control group is included; the sequential conduct of multiple assessments further precludes attribution of delayed PEM to individual procedures, limiting analyses to feasibility and tolerability at the level of assessment batteries.

INTRODUCTION

Background and rationale

The clinical picture of myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) is defined as a severe, chronic, multisystem disease. Depending on its severity, it can lead to substantial physical and cognitive impairments—potentially resulting in incapacity to work, the need for long-term care, artificial nutrition or, in extreme cases, death.[1 2] Among the systems affected are the central and autonomic nervous systems, the immune system, the cardiovascular system (particularly endothelial cells), and mitochondrial energy metabolism. Common symptoms include orthostatic intolerance with dizziness, tachycardia or syncope on standing, as well as disturbances in blood circulation, the gut microbiome and nerve fibres. Reactivations of persistent viruses, such as herpesviruses, may also contribute to further physiological damage.[2–5]

In approximately 80% of cases, ME/CFS develops following an infectious illness, commonly triggered by viruses such as Epstein-Barr virus, influenza virus or SARS-CoV-2.[2 6 7] Surgical procedures, resuscitations and head or neck trauma are also recognised as potential triggers; in some cases, however, the underlying cause remains unknown.[4]

The onset of the disease is often insidious or episodic.[2] The hallmark symptom is post-exertional malaise (PEM)—a profound worsening of overall health following minimal exertion, accompanied by muscle pain, fatigue and flu-like symptoms.[8] Other common symptoms include dizziness, sleep disturbances, cognitive impairments, pain syndromes, temperature dysregulation and

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marked hypersensitivity to stimuli and food.[9] Approximately 25% of affected individuals are largely housebound, and over 60% are unable to work.[10]

Prior to the pandemic, the global prevalence was estimated at 0.3%–0.9%. In Germany, according to the IQWiG, there were between 140 000 and 310 000 cases, including 70 000 to 90 000 children and adolescents.[11 12] According to studies, it is assumed that the number of affected individuals has at least doubled since the SARS-CoV-2 pandemic.[4 13] According to a statement by the National Association of Statutory Health Insurance Physicians (Kassenärztliche Bundesvereinigung), the overall prevalence of ME/CFS in Germany in 2021 was approximately 500 000 cases.[14]

The disease pattern can be categorised into different groups based on the severity of the disease:

- Mild ME/CFS: despite burdensome symptoms (eg, pain, fatigue, sleep disturbances), employment remains possible.
- Moderate ME/CFS: work capacity is no longer preserved; daily activities are limited and require frequent rest periods.
- Severe ME/CFS: only minimal activities, such as eating or personal hygiene, remain possible. Many affected individuals require a wheelchair and recover only slowly from exertion.
- Very severe ME/CFS: affected individuals are bedridden, entirely dependent on assistance and highly sensitive to light and sound.[15]

The prognosis for individuals with severe and very severe ME/CFS is significantly poorer. To date, no specific or effective treatment is available.[16] Although specialised ME/CFS services within the NHS report improvements in individual symptoms such as fatigue, anxiety or depression through cognitive behavioural therapy and graded exercise therapy, no distinctions were made regarding disease severity.[17]

A central limitation of the current body of research is the insufficient inclusion of patients with severe and very severe ME/CFS into clinical studies. This patient group is often excluded from studies—primarily due to immobility, lack of caregiver support and pronounced exertional intolerance, with the risk of symptom exacerbation caused by PEM. Cox and Findley[15] reported that >60% of severe ME/CFS patients were unable to complete inpatient diagnostic protocols due to PEM, underscoring the need for adaptive, home-based approaches.[1 15]

While several recent studies have employed low-burden diagnostic approaches—such as the analysis of blood-based parameters or the use of various standardised questionnaires[18 19]—there remains a lack of diagnostic methods that are both minimally taxing and capable of accurately assessing key systems affected in ME/CFS. These include, in particular, the central and autonomic nervous systems, as well as the cardiovascular system, especially endothelial function.[20]

While blood biomarkers (eg, cytokine panels) and questionnaires (eg, Questionnaires for symptom assessment as[18 19]) show promise, they often fail to capture autonomic dysfunction (eg, heart rate variability (HRV), pupillography) or endothelial impairment (eg, non-invasive perfusion assessments as[20]) in severe ME/CFS due to the burden of clinic visits. On the other hand, autonomic abnormalities[18] and endothelial dysfunction[21] are increasingly recognised in ME/CFS, their assessment in severe patients is hampered by the risk of PEM[8 15] and logistical barriers.[15] Furthermore, when assessing cognitive impairments, diagnostic equipment is more suitable than assessment using questionnaires. Home-based alternatives for these diagnostic tools are urgently needed for several reasons: Identifying tolerable diagnostic tools could **accelerate enrolment in clinical trials** (eg, for antivirals or immunoadsorption) by reducing dropout rates. Home-based assessments may reduce noise from clinic-induced stress, providing more **ecologically valid** data. Severely affected patients are often **excluded from research** due to logistical barriers and because of the potential risk to harm them, especially with the risk of PEM. This study aims to determine the tolerability of various test procedures and thus contribute to patient safety in future studies.

It is crucial that such diagnostic procedures are tolerable for severely affected ME/CFS patients. Therefore, a systematic investigation of this patient subgroup is urgently needed to develop more targeted diagnostic, therapeutic and care strategies. Once therapeutic options are available, repeated examinations of different parameters are necessary to objectify a response to treatment.

Objectives

Primary objective

The primary objective of this study is to evaluate the feasibility and tolerability of home-based clinical and instrumental assessments in patients with severe to very severe ME/CFS (Bell score ≤30). Specifically, we aim to:

1. Determine the proportion of assessments that can be completed without triggering PEM or requiring premature termination.
2. Identify patient-reported and physiological barriers (eg, fatigue, pain, cognitive dysfunction) to completing standardised diagnostic procedures in this population.

Secondary objectives

1. Stratify feasibility by assessment type:
   - Compare completion rates between low-burden (eg, resting ECG, pupillography), moderate-burden (eg, HRV with paced breathing) and high-burden (eg, cognitive tests, Endopat-test) procedures.
   - Assess whether **supine-position tests** are better tolerated than seated/standing tests.
2. Explore associations between disease severity and tolerability:
   - Examine whether participants with Bell scores ≤20 (very severe) have lower completion rates than those with scores 21–30 (severe).

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Table 1 Questionnaires phase 1

Questionnaires	Content
Composite autonomic symptom score-31	► Evaluation of autonomic symptom burden in patients ► Assessment of symptoms indicative of autonomic nervous system dysfunction, including dizziness, palpitations, gastrointestinal disturbances and related complaints
Short form 12	► Assessment of overall quality of life and general health ► Items addressing physical and mental health domains, including general health status, physical functioning, social functioning and emotional well-being
Fatigue life experience inventory	Assessment of fatigue and its impact on daily functioning
Munich Berlin symptom questionnaire	► Assessment of symptoms associated with myalgic encephalomyelitis/chronic fatigue syndrome ► Documentation of the frequency, severity and duration of core symptoms, such as fatigue, post-exertional malaise (PEM), cognitive impairment, sleep disturbances and pain
DePaul symptom questionnaire–PEM	► Assessment of PEM ► Evaluation of the type, severity, frequency and delayed onset of PEM following physical or cognitive exertion

- Investigate whether disease duration or specific symptom clusters (eg, orthostatic intolerance, cognitive dysfunction) predict assessment tolerability.

3. Establish baseline values for home-based biomarkers:
   - Document descriptive distributions of selected biomarkers (eg, HRV, cortisol levels, endothelial function) within a cohort of patients with severe ME/CFS, providing a data basis for future controlled studies.
4. Pilot a biobank for ME/CFS research:
   - Collect and store blood and saliva samples to enable future investigations into biomarkers (eg, viral reactivation, autoantibodies) and therapeutic targets.

Hypotheses

Primary Hypotheses

- **H1**: fewer than 50% of standard clinical assessments (eg, cognitive tests, orthostatic challenges) will be fully completable in patients with severe ME/CFS (Bell score ≤30) due to PEM or physical/cognitive exhaustion.
  - *Rationale*: severe ME/CFS is characterised by exertion intolerance, and even minimal activity (eg, seated cognitive tests) can trigger PEM.[2 8] Hospital-based studies report high dropout rates in this subgroup.[15]

Secondary hypotheses

- **H2**: non-invasive, supine-position assessments (eg, resting ECG, pupillography, HRV) will have ≥80% completion rates, while high-burden tests (eg, Trail Making Test-B, standing tests) will have <30% completion rates.
  - *Rationale*: supine tests minimise orthostatic stress and physical exertion, reducing PEM risk.[21]
- **H3**: participants with very severe ME/CFS (Bell score ≤20) will tolerate fewer assessments than those with severe ME/CFS (Bell score 21–30), with a ≥20% difference in completion rates.
  - *Rationale*: very severe patients exhibit greater sensitivity to exertion and often require total bed rest.[15 22]
- **H4**: autonomic (eg, HRV, pupillary responses) and endothelial parameters (eg, peripheral perfusion) are expected to show deviations from device-specific reference values in a substantial proportion of participants.
  - *Rationale*: prior studies link ME/CFS to autonomic and vascular dysfunction,[21] but home-based validation is lacking.

Exploratory aims

In addition to the primary and secondary objectives, this study will explore:

1. Correlations between PEM severity (post-assessment) and baseline biomarkers (eg, HRV, cortisol).
2. Patient-reported acceptability of home-based assessments via Likert-scale surveys (eg, "How tolerable was this test?").
3. Logistical challenges (eg, caregiver availability, equipment usability) to inform scalability of home-based diagnostics in future trials.

METHODS

Study design

The study is a prospective, non-interventional observational study with a single-group design and an exploratory framework. Participant recruitment began in September 2025, and the study is expected to be completed by the end of April 2026.

Study setting

The study is based at the Institute for Exercise and Occupational Medicine at the University Medical Centre Freiburg and is conducted as an outreach study. Data collection takes place in the patients' home environment through home visits and questionnaires.

Participants

Eligible participants include male and female patients aged 18 and older with severe to very severe ME/CFS (Bell score ≤30). Additionally, it is required that patients

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Table 2 Assessments phase 2a

Assessments	Content	Expected burden	Duration
Resting ECG using a 12-lead custo cardio 100 ECG device (custo med GmbH, Ottobrunn, Germany)	► Measurement of the heart's electrical activity ► Assessment of cardiac function ► Exclusion of various cardiac diseases	Low (minimal exertion)	5 min
Measurement of **heart rate variability** using a 12-lead custo cardio 100 ECG device (custo med GmbH, Ottobrunn, Deutschland) and Kubios scientific software V.4.1 (Kubios, Kuopio, Finland)	Assessment of beat-to-beat heart rate fluctuations over a short period (5 min) using ECG, following several cycles of inhalation and exhalation	Moderate (requires controlled breathing)	15 min
Pupillography using the SomnoMeter F^2D^2 device (AMTech Pupilknowlogy GmbH, Dossenheim, Germany)	► Objective measurement and analysis of central nervous system activation ► Assessment of pupillary response to provide indications of daytime sleepiness	Low	11 min
Application of the **Light Move four activity sensor** (10401) (movisens GmbH, Karlsruhe, Germany)	Assessment of sleep quality, physical activity, sunlight exposure, step count and energy expenditure over a period of 3 days	Low	2 min
Cognitive assessments ► MoCA test (Montreal cognitive assessment) (MoCA test, Québec, Canada) ► TMT-B (trail making test) (Schuhfried, Mödling, Österreich) ► SDMT test (Symbol digit modalities test) (Hogrefe, Oxford, UK)	► Assessments of memory retention, sustained attention and concentration performance ► Assessment of memory, attention, language, executive functions and visuospatial abilities ► Assessment of the ability to switch between tasks and to perform complex visual tasks quickly and accurately ► Assessment of how quickly an individual can associate symbols with numbers	High (mental exertion)	30 min
Osteosonography using the Sunlight MiniOmni device (software V.4.2) (MICRO-MEDICAL Instrumente GmbH, Senftenberg, Germany)	► Assessment of bone mineral density ► Evaluation of the risk for osteoporosis and bone fractures	Low	5 min
Diaphragm ultrasonography using the Lumifiy S4-1 (Koninklijke Philips N.V., Amsterdam, the Netherlands)	Assessment of diaphragmatic function and movement	Moderate	5 min
Assessment of body composition using Bioelectrical Impedance Analysis (Data Input GmbH, Pöcking, Germany)	Simple and precise method for assessing body composition	Low	5 min
Blood sampling	► Exclusion of other underlying diseases ► Assessment of: thyroid function (thyroid-stimulating hormone (TSH)), renal function (creatinine), ferritin, high-sensitivity C-reactive protein to investigate possible anaemia, vitamin D, SARS-CoV-2 antibodies, Epstein-Barr virus and cytomegalovirus (approximately 30 mL)	Moderate (invasive)	5 min
Saliva sample	Assessment of cortisol levels (measured before and after the assessments)	Moderate	2 min

find the planned diagnostic procedures acceptable. Only patients living within a maximum distance of 50 km from Freiburg will be included. Exclusion criteria include absolute arrhythmia, blindness or amelia. All patients must provide written informed consent for participation and pseudonymised data use before the study begins.

Variables

The primary variables of interest were feasibility and tolerability of the planned home-based assessments. Feasibility was defined as the proportion of assessments completed with the receipt of valid measured values, while tolerability was assessed based on premature termination, need for additional rest periods and patient-reported PEM following assessments. Outcomes are evaluated per individual assessment and documented at predefined follow-up time points after each study visit. Secondary variables included autonomic, cognitive, vascular, activity-related and biological parameters collected during home visits. Disease-related variables such as disease severity (Bell Score), disease duration and symptom burden were recorded as contextual variables. Potential confounders included age, sex, medication use and testing conditions.

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Table 3 Assessments phase 2b

Assessments	Content	Expected burden	Duration
Near-infrared spectroscopy measurement of muscle tissue using Moxy (TRAINALYZED GmbH; Potsdam, Germany)	Assessment of muscle oxygenation and blood flow	Low	3 min
Neurocognitive/psychological assessment using the Vistec app (PC software V.3.0.004) (Vistec AG, Olching, Germany)	► Tests of memory performance, concentration and attention ► Assessment of: – Intrinsic alertness – Sustained attention (short version)	High (Mental exertion)	30 min
Measurement of handgrip strength using the Jamar Plus dynamometer (2019070417) (Performance Health Supply, Illinois, US)	Assessment of the combined strength of the muscles in the hands, wrists and forearms	Moderate	3 min
Schellong test	Assessment of the body's orthostatic response, particularly the regulation of blood pressure and heart rate during postural changes	Moderate (standing/sitting)	15 min
Assessment of endothelial function using EndoPATx (OM1695230) (Itamar medical; Caesarea, Israel)	Assessment of endothelial function, the inner cellular lining of blood vessels	High (ischaemic pain)	25 min
Saliva sample	Assessment of cortisol levels (measured before and after the assessments)	Moderate	2 min

Data sources/measurement

Phase 1

The study is advertised via press releases, newspapers, websites and through contact with support groups. Comprehensive patient information for Phases 1 and 2 is provided on the website of the Institute of Occupational and Exercise Medicine. Interested individuals are encouraged to make contact via email or phone. Participants are informed about the study purpose and procedures via e-mail and receive the patient information and informed consent form. After all inclusion criteria are confirmed and the signed informed consent is received, the questionnaires for Phase 1 of the study are sent. During any further surveys or assessments, a caregiver may be present if needed. The first survey includes the following topics:

- Diagnostic criteria for ME/CFS (Canadian consensus criteria).
- Severity of ME/CFS-related impairments (Bell score).
- Care level, if applicable.

All interested individuals who meet the inclusion criteria (Bell score ≤30; care level 3–5) and provide written consent will be interviewed via (video) call using a structured physician questionnaire, covering the following:

- Duration of illness, prior medical evaluations, current medication.
- Subjective assessment of daily stress and exertion tolerance, particularly thresholds for crash symptoms.
- Willingness to proceed with further diagnostic phases and logistical feasibility.

Participants identified as having severe to very severe ME/CFS and who provide initial consent will receive additional questionnaires (online or by mail). Online questionnaires are administered via the REDCap system and follow the same process as the initial survey.

Table 1 provides an overview of the corresponding questionnaires.

All questionnaires should be completed and returned to the institute within two weeks. Non-responders will receive one reminder; delays >4 weeks will lead to exclusion. The aim of the survey is to comprehensively and accurately assess the individual burden of illness based on self-reported information. One week after sending out the questionnaires, follow-up contact is made to assess feasibility and tolerability. Additionally, participation in the outreach-based Phase 2 is discussed.

Phase 2

Phase 2 involves various non-invasive assessments and blood sampling, covering autonomic, cognitive and metabolic domains. A staged approach (two sub-phases (2a and 2b), conducted through home visits) is intended to minimise the burden for patients.

Phase 2a

Participants who return a signed consent form for Phase 2 will be contacted again by mail to schedule home visits.

Table 2 presents the planned assessments in the listed order.

The assessments, listed in table 2, are scheduled in an order that places physically demanding or stress-sensitive procedures last. If necessary, the examinations will be carried out in the presence of the caregiver. If patients are advised against standing tests due to PEM risk, we will perform tests with a supine-position prioritisation.

One week after the assessments, participants will be contacted again and will receive a questionnaire to assess PEM.[23] In addition, participants will be asked whether they are willing to take part in further assessments. If

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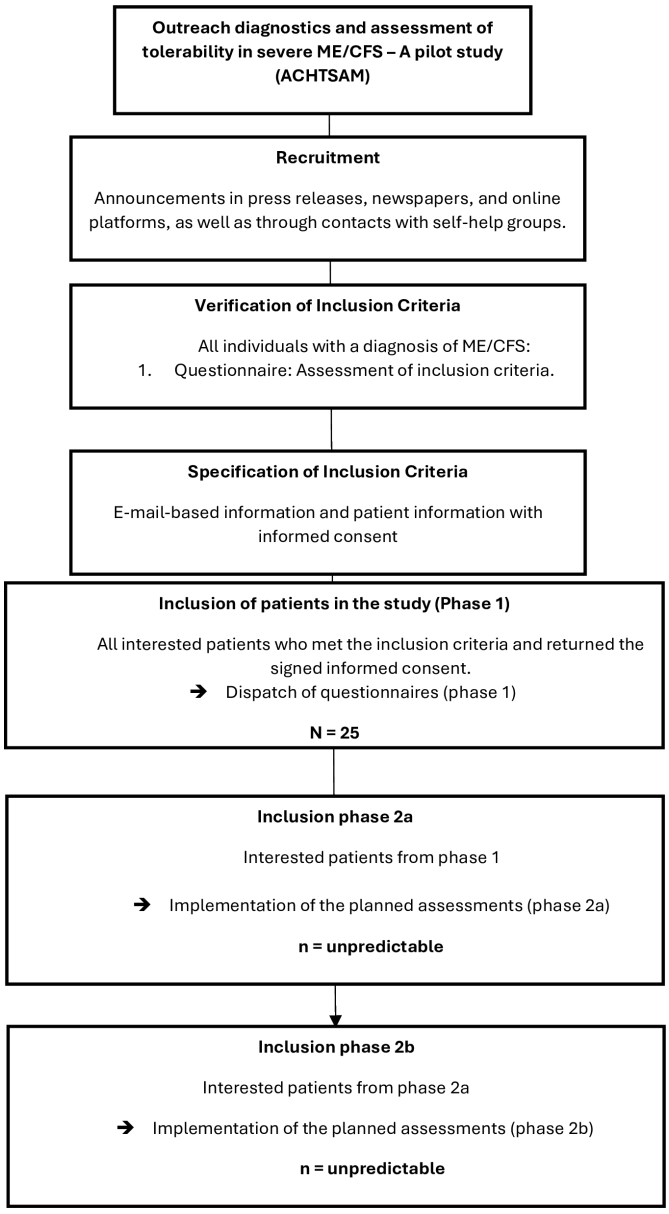

Figure 1 Timeline of patient enrolment and study phases of the ACHTSAM pilot study, illustrating the sequence of phases. ME/CFS, myalgic encephalomyelitis/chronic fatigue syndrome.

appropriate consent is provided and this is considered reasonable, a follow-up visit will be scheduled.

There is a minimum 2 week interval between Phases 2a and 2b.

Phase 2b

Phase 2b includes additional non-invasive assessments, listed in table 3, conducted through home visits.

One week after these assessments, participants will again be contacted and provided with a questionnaire to report any worsening of PEM. Documentation includes which procedures were completed and any issues encountered during or after testing (eg, fatigue or physical decline).

Data collection

The selection of the assessment methods and measurement devices used in Phase 2 was based on the prerequisite that all devices hold valid certification as medical devices. This regulatory classification implies standardised testing with regard to safety, functionality and measurement accuracy, allowing for the assumption of high validity and reliability of the collected data. A methodological exception to this requirement is made for the procedures used to assess HRV and endothelial function.

The HRV measurement is a non-invasive and painless measurement method that has been previously employed in studies involving conditions associated with dysregulation of the autonomic nervous system and elevated stress levels (eg, musculoskeletal pain).[24] Given that similar autonomic dysfunctions have been described in ME/CFS, the use of HRV measurement in this study is deemed appropriate for detecting potential pathophysiological changes.

Endothelial function is assessed using the EndoPAT device, which enables non-invasive peripheral arterial tonometry at the fingertip. Evidence of vascular dysfunction and impaired vascular regulation in patients with ME/CFS suggests that this method is suitable for investigation in this context. Accordingly, it has already been applied in previous studies involving individuals with ME/CFS.[21 25]

Participant timeline

The inclusion of 25 patients with severe to very severe ME/CFS is planned within a 3 month period. The timeline of patient enrolment and study phases is shown in figure 1.

Bias

Potential sources of bias include selection bias, as participation is limited to patients with severe or very severe ME/CFS who are able to tolerate home-based assessments and provide informed consent. Consequently, individuals with extremely severe disease or insufficient caregiver support may be underrepresented.

Attrition bias may occur due to premature termination of individual assessments as a result of symptom exacerbation. To address this, all incomplete assessments and the reasons for their termination are systematically documented.

Information and measurement bias may arise from symptom fluctuations and fatigue-related performance variability. To minimise this risk, assessments are conducted in a staged approach and may be paused or discontinued if necessary.

Sample size justification

A sample of 25 is sufficient to estimate feasibility with 95% CIs ±20% (assuming 50% completion rate). This aligns with pilot study guidelines.[26 27] Subgroup analyses (eg, by Bell score tertiles) will be exploratory due to limited power.

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Quantitative variables

Disease severity will be analysed both as a continuous variable and categorised according to predefined Bell score thresholds for exploratory analyses.

Statistical methods

The descriptive analysis of the feasibility of assessments in patients with severe ME/CFS will initially be conducted for each type of examination separately. This will involve calculating the proportion of assessments that could be completed in full and yielded valid results. Specifically, the number of successfully completed tests will be expressed as a proportion of the total number of planned tests. In addition, feasibility will be compared across subgroups based on age, sex, disease severity and functional capacity.

Furthermore, the feasibility of study-related procedures will be analysed in clusters (eg, questionnaires, diagnostic measures performed in supine/sitting/standing positions, neurocognitive tests, perfusion assessments, etc). This approach will help identify which types of assessments can reasonably be administered to this patient population and which may be too burdensome.

Subgroup comparisons (eg, Bell score ≤20 vs 21–30) will use Fisher's exact tests for categorical outcomes.

The identification and evaluation of challenges in conducting clinical assessments will be performed using descriptive statistics. Reasons for test failure or limitations in test administration will be recorded, and the frequencies of various causes of limitations will be analysed.

Tolerability of individual assessments and clusters will also be evaluated. This includes determining whether additional rest periods were required after specific tests, whether assessments had to be terminated prematurely, or whether symptoms occurred subsequently.

The study is not powered to estimate prevalence, validate biomarkers or establish normative reference values in this specific group of patients. All biological outcomes will be analysed descriptively and interpreted exclusively in an exploratory context.

Data management

The data collected in the context of this study will be stored both electronically and in paper form at the Universitätsklinikum Freiburg and processed in accordance with the general data protection regulation. All data will be pseudonymised, meaning that personal identifiers such as names or addresses will be replaced by a code number. A securely stored re-identification list will be accessible only to authorised personnel. Online and sensor data will likewise be pseudonymised and securely stored. Biological samples will be encrypted and stored for 3 years in the central biobank. Any publications will be based solely on anonymised data. Contact with treating physicians will occur only with written informed consent. The study management team is responsible for data protection.

Harms

Participation in this study is intended for the scientific investigation of the physical and cognitive capacity of individuals with ME/CFS and aims to contribute to the long-term improvement of diagnostic and therapeutic approaches. No direct medical benefit for participants is expected. Potential risks are associated with participation, including local reactions to blood sampling as well as possible cognitive, emotional or physical overexertion. In particular, there is a risk of PEM, a characteristic worsening of symptoms following exertion, which may be prolonged or, in rare cases, irreversible. To minimise these risks, an individualised assessment of exertion tolerance is performed in advance, all assessments are carried out in a staged and symptom-oriented manner, and participants are continuously monitored. Examinations may be paused or discontinued at any time. Rest periods between examinations are given as needed. Symptom tracking (ie, questions regarding comfort and well-being) is done during the evaluation steps. In addition, participants will be contacted one week after the assessments to determine whether they experienced PEM symptoms. Participation is also possible in a reduced form. The study procedures can be interrupted or terminated at any time, and informed consent may be withdrawn without any disadvantage to the participant.

Patient and public involvement

The study protocol was developed in collaboration with the ME/CFS Freiburg Initiative, two individuals affected by ME/CFS, and one caregiver.

ETHICS AND DISSEMINATION

The study protocol was approved by the Ethics Committee of the University of Freiburg (No. 25-1031-S1). Written informed consent is obtained from all participants prior to study inclusion. Given the vulnerability of the study population, assessments are conducted in a staged approach, with the option to pause or terminate individual procedures at any time in order to minimise the risk of PEM.

Study results will be disseminated through peer-reviewed publications and shared with patient support groups. Pseudonymised data will be made available on reasonable request to the corresponding author in accordance with data protection regulations.

Contributors PD and WK designed the study. CF coordinated data collection and drafted the manuscript. PD provided methodological supervision. All authors critically revised the manuscript and approved the final version. CF is the guarantor.

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Competing interests None declared.

Patient and public involvement Patients and/or the public were involved in the design, or conduct, or reporting, or dissemination plans of this research. Refer to the Methods section for further details.

Patient consent for publication Not applicable.

Ethics approval This study involves human participants and was approved by the Ethics Committee of the University of Freiburg (No. 25-1031-S1). Written informed

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consent is obtained from all participants. Results will be disseminated via peer-reviewed publications and patient support groups. Participants gave informed consent to participate in the study before taking part.

Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement Data are available upon reasonable request.

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